

Topi Halme

Postdoctoral Researcher, Aalto University

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Research Interests

Sequential inference, quickest change detection, distributed inference, statistical signal processing, multiple hypothesis testing, empirical Bayes methods, conformal inference, game-theoretic probability/statistics, and causal inference.

Experience

- 2025-** Postdoctoral Researcher, Aalto University.
2021–2025 Doctoral Researcher, Aalto University.

Education

- 2022–2025** PhD in Electrical Engineering, Aalto University.
Thesis: “*Detecting changes in Large-Scale Streaming Data*”
2019–2022 M.Sc. in Statistics, University of Helsinki.
Thesis: “*Quickest Detection under False Discovery Rate and Communication Constraints*” (grade 5/5)
2015–2019 B.Soc.Sc. in Statistics, University of Helsinki.

Publications

Preprints

- [P1] Topi Halme and V. Koivunen. “Multi-stream Quickest Change Detection: Foundations and Recent Advances”. Submitted to *Entropy*. 2026. arXiv: 2604.18008.
[P2] J. Poropudas and Topi Halme. “Dean Oliver’s Four Factors Revisited”. Submitted to *Journal of Quantitative Analysis in Sports*. 2026. arXiv: 2305.13032.

Journals

- [J1] Topi Halme, V. V. Veeravalli, and V. Koivunen. “Quickest change detection for multiple data streams using the James-Stein estimator”. In: *IEEE Transactions on Information Theory* (2025). arXiv: 2404.05486.
[J2] Topi Halme, E. Nitzan, and V. Koivunen. “Bayesian quickest detection of propagating spatial events”. In: *IEEE Transactions on Signal Processing* (2022).
[J3] E. Nitzan, Topi Halme, and V. Koivunen. “Bayesian methods for multiple change-point detection with reduced communication”. In: *IEEE Transactions on Signal Processing* (2020). arXiv: 2003.11062.

Peer-reviewed conferences

- [C1] Topi Halme, Venugopal V Veeravalli, and Visa Koivunen. “Quickest Change Detection of Unknown Mean-Shifts using the James-Stein Estimator”. In: *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2025.

- [C2] Topi Halme and Visa Koivunen. “Optimal Multi-Stream Quickest Detection with False Discovery Rate Control”. In: *2023 57th Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2023.
- [C3] Topi Halme, Eyal Nitzan, and Visa Koivunen. “Bayesian multiple change-point detection of propagating events”. In: *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2021.
- [C4] Topi Halme, Eyal Nitzan, and Visa Koivunen. “Quickest detection of propagating signals for flexible radio spectrum use”. In: *2021 55th Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2021.
- [C5] Topi Halme, Eyal Nitzan, H. Vincent Poor, and Visa Koivunen. “Bayesian multiple change-point detection with limited communication”. In: *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2020.
- [C6] Eyal Nitzan, Topi Halme, H. Vincent Poor, and Visa Koivunen. “Deterministic multiple change-point detection with limited communication”. In: *2020 54th Annual Conference on Information Sciences and Systems (CISS)*. IEEE. 2020.
- [C7] Martin Gözl, Michael Muma, Topi Halme, Abdelhak Zoubir, and Visa Koivunen. “Spatial inference in sensor networks using multiple hypothesis testing and bayesian clustering”. In: *2019 27th European Signal Processing Conference (EUSIPCO)*. IEEE. 2019.
- [C8] Topi Halme. “Bayesian Multiple Hypothesis Testing For Distributed Detection In Sensor Networks”. In: *IEEE Data Science Workshop*. 2019.
- [C9] Topi Halme and Visa Koivunen. “Nonparametric Distributed Detection Using One-Sample Anderson-Darling Test and p-value Fusion”. In: *IEEE Data Science Workshop*. 2018.
- [C10] Topi Halme, Visa Koivunen, and H. Vincent Poor. “Nonparametric distributed detection using bootstrapping and Fisher’s method”. In: *2018 52nd Annual Conference on Information Sciences and Systems (CISS)*. IEEE. 2018.

Theses

- [T1] Topi Halme. “Detecting changes in large scale streaming data”. PhD thesis. Aalto University, 2025.
- [T2] Topi Halme. “Quickest Detection under False Discovery Rate and Communication Constraints”. MA thesis. Aalto University, 2021.

Awards

- 2025** Aalto ELEC thesis award (awarded to top 10% of theses in the department)
- 2023** Nokia Foundation Scholarship
- 2021–2022** Leo Törnqvist Award (best Master’s thesis in statistics in Finland)

Teaching

Guest lecturer

- 2026** ELEC-5431 Large-Scale Data Analysis

Teaching assistant

- 2023** ELEC-E5440 Statistical Signal Processing
- 2024** ELEC-E5440 Statistical Signal Processing

2025 ELEC-E5440 Statistical Signal Processing

Reviewing

IEEE TSP, IEEE TIT, ISIT